SIMBLOCKS // RESULT

2026-10-15 22:34Z

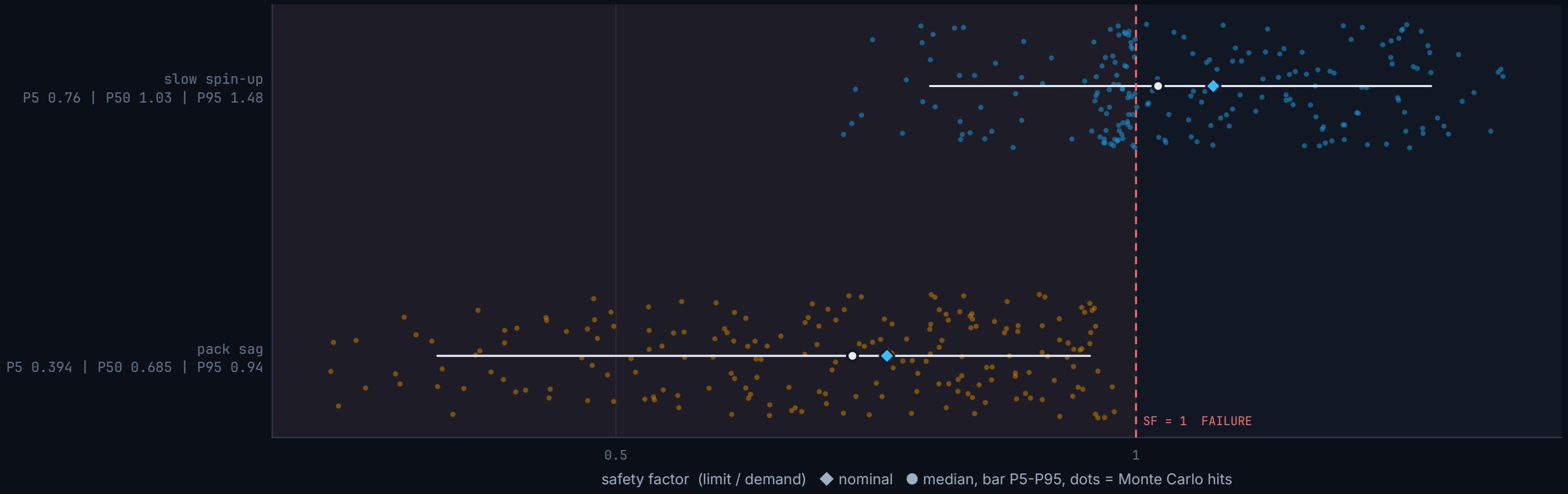
TRIAL 1: WEAPON SPIN-UP

● PASS VERIFIED

How fast does the weapon spin up, and does the pack sag below 3.3 V/cell?

WEAPON SETUP OK?	SLOW SPIN-UP	PACK SAG [GOVERNING]
FAIL	WARN	FAIL
NO	SF 1.11	SF 0.718
95% speed in 1.80 s, pack min 2.37 V/cell	P(fail) 46% | SF P5 0.76	P(fail) 100% | SF P5 0.394

Margin to each user-defined failure mode



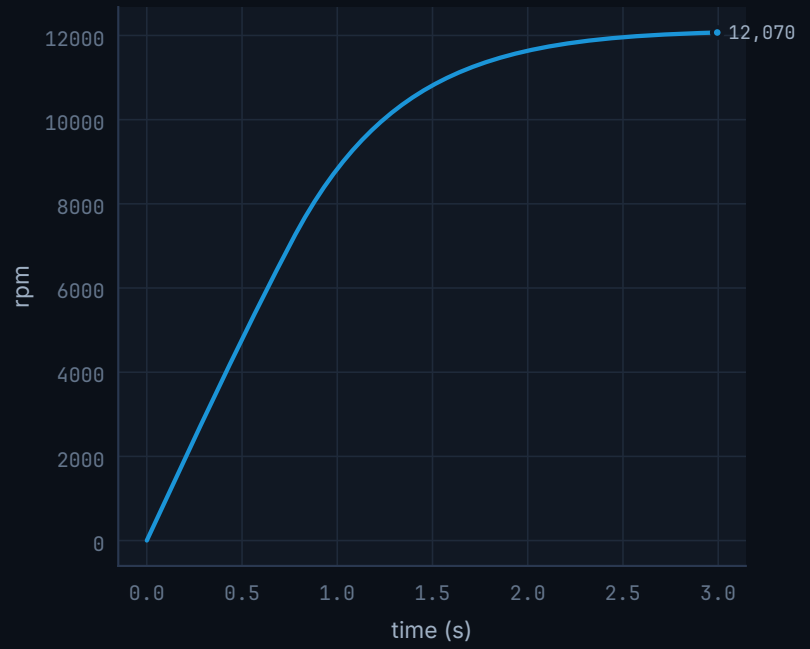
Trial 1 weapon spin-up: VERIFIED | PASS 13 WARN 2 FAIL 0 SKIP 0 | 222 runs, 2.0 s

2026-10-15 22:34Z

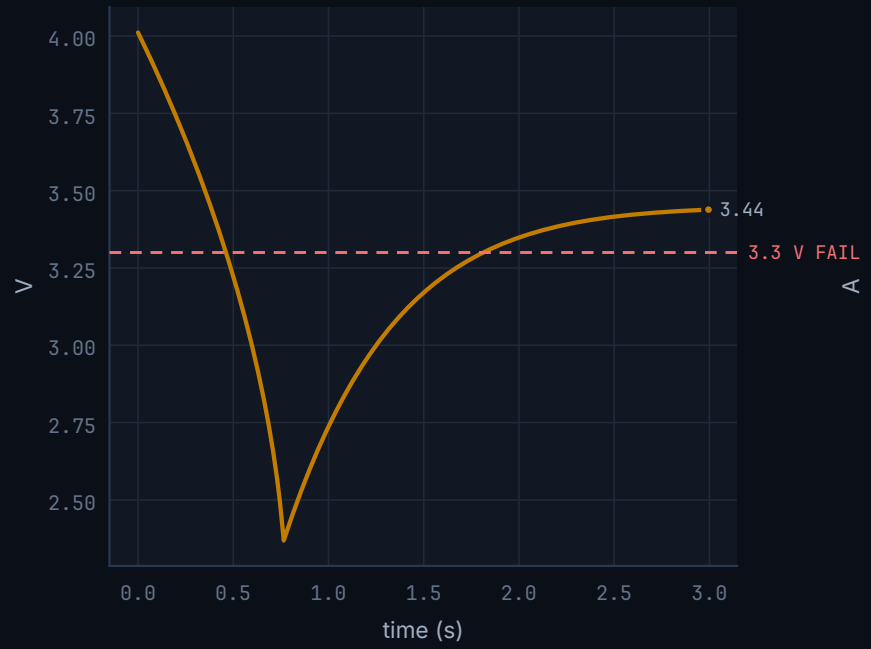
TRIAL 1: WEAPON SPIN-UP

Time histories of the governing states (nominal case)

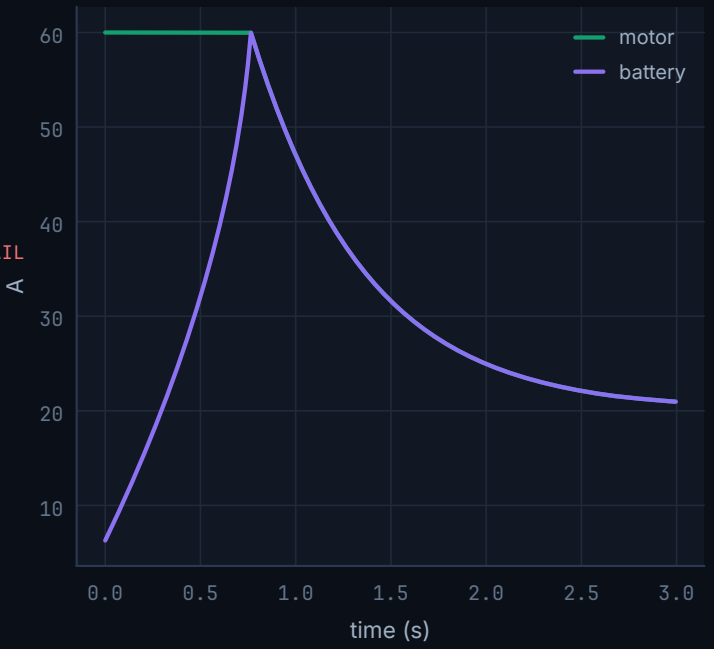
Weapon speed



Pack voltage per cell



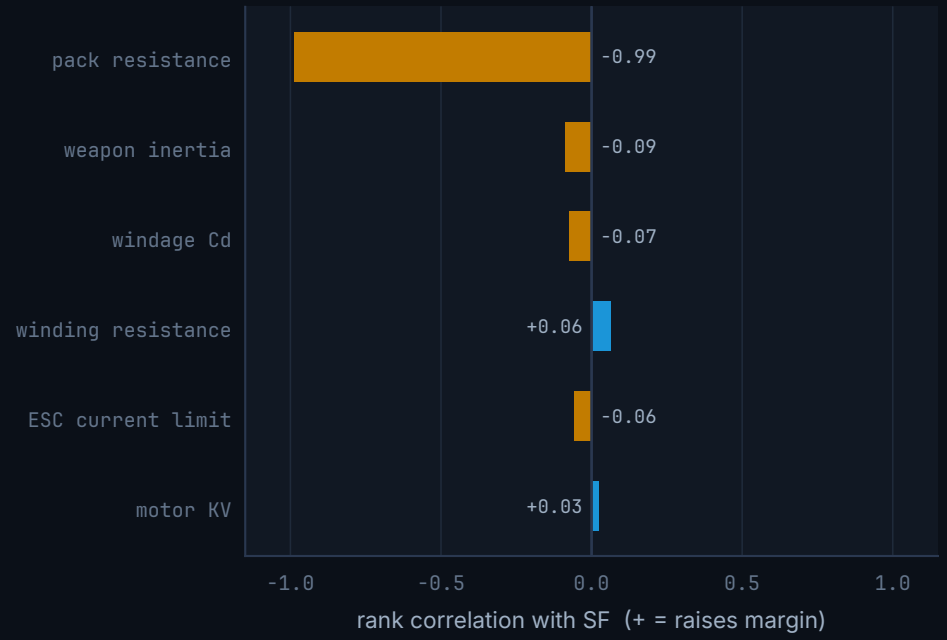
Currents



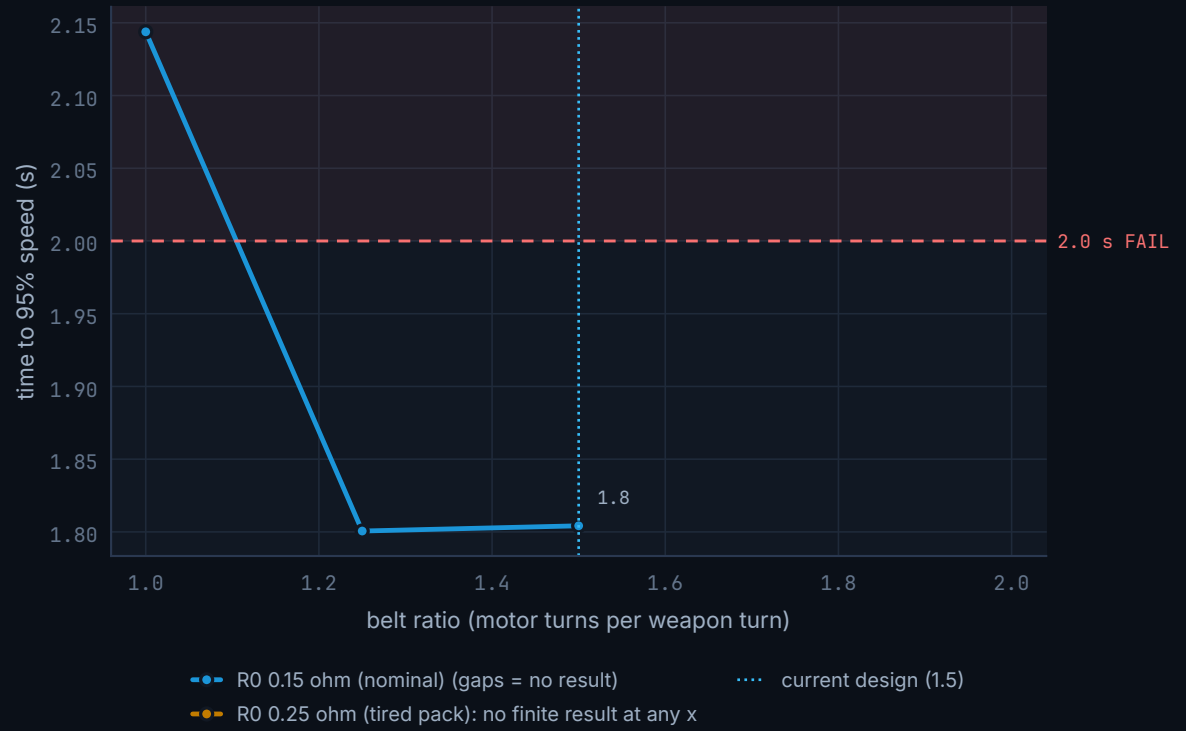
TRIAL 1: WEAPON SPIN-UP

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'pack sag' margin



Spin-up time vs belt ratio



SIMBLOCKS // VERIFICATION

2026-10-15 22:34Z

TRIAL 1: WEAPON SPIN-UP

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 0.316 s / dt 0.0001 s = 3156.0x
dt_convergence	● WARN	t95_s: err 0.0055% (order inf); rpm_final: err 0.00037% (order 1); v_min_per_cell: err 0.0042% (order 5.7); I_peak_A: err 0% (order inf); mAh_used:...
energy_ledger	● PASS	worst closure residual 0.002% across 3 dt levels (tol 5%)
limit:ESC limit 0 A → weapon never moves	● PASS	matches expected limit
limit:ideal pack, no losses → spin_up_time block	● PASS	matches expected limit
knockout:no windage	● PASS	t95_s -24.4%, rpm_final +20.3%, v_min_per_cell +0.0763%, I_peak_A +0%, mAh_used -26.5%
knockout:no bearing drag	● PASS	t95_s -0.61%, rpm_final +0.129%, v_min_per_cell +0.00249%, I_peak_A +0%, mAh_used -0.222%
knockout:no RC polarization	● PASS	t95_s -1.26%, rpm_final +0.495%, v_min_per_cell +0.278%, I_peak_A +0%, mAh_used +0.628%
knockout:no pack sag	● PASS	t95_s -27.1%, rpm_final +18.3%, v_min_per_cell +72.6%, I_peak_A +0%, mAh_used +13.3%
sensitivity	● PASS	6 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	200 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	slow spin-up: nominal SF 1.11, P(fail) 46.5%; pack sag: nominal SF 0.718 FAILED, P(fail) 100%